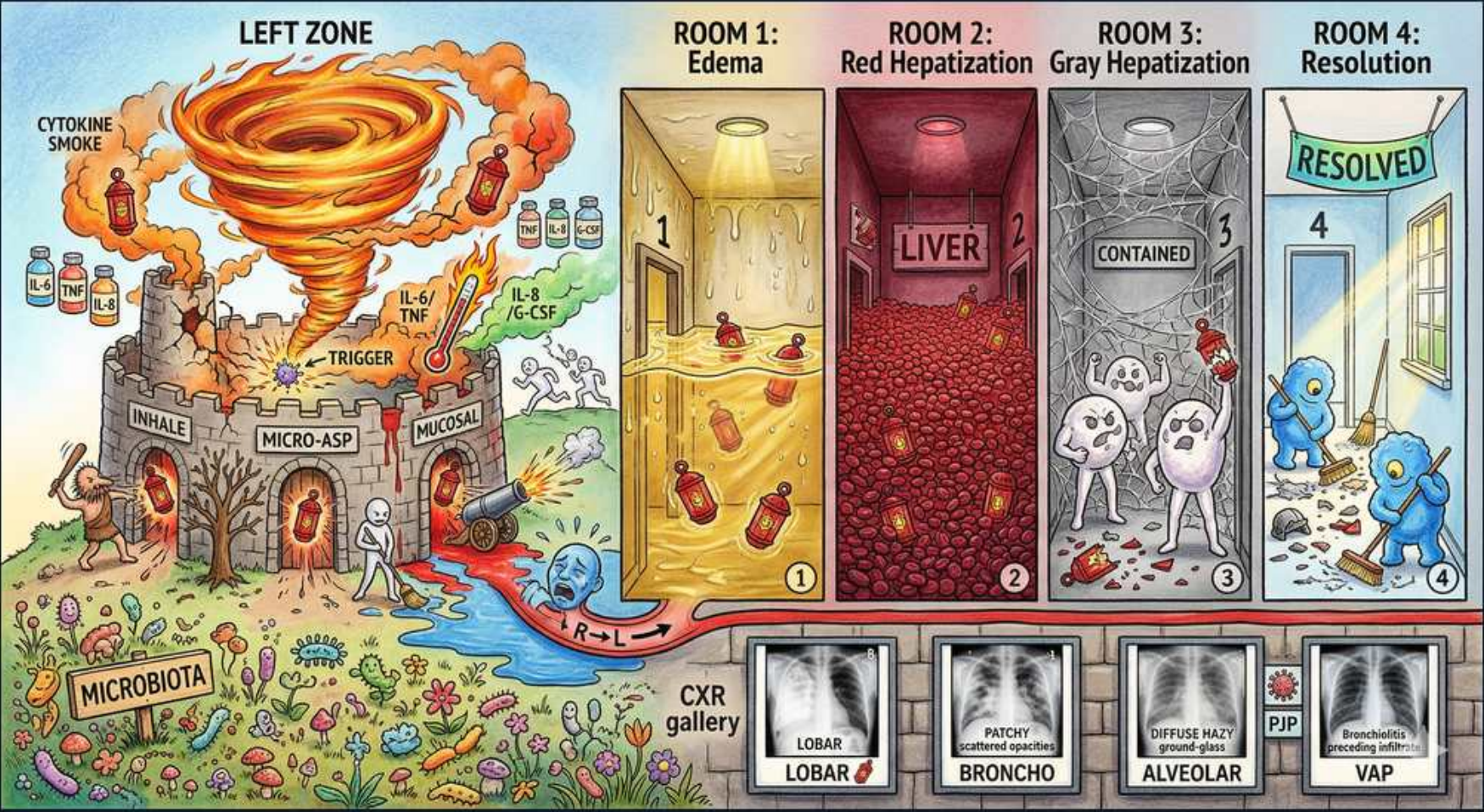


Pneumonia — Pathogenesis & Phases of Lobar Consolidation



1. Microbiota / colonization

WHAT YOU SEE

Flower-bed 'microbiota' at castle base

WHAT IT ENCODES

Pneumonia begins when normal upper-airway flora is disrupted and pathogens colonize. Streptococcus pneumoniae is the leading cause of community-acquired pneumonia. Colonization precedes invasion.

BOARD TRAP

Colonization alone is not infection — sputum culture growth without clinical signs can reflect colonization, especially with Gram-negatives in hospitalized patients.

2. Micro-aspiration

WHAT YOU SEE

'MICRO-ASP' on castle wall

WHAT IT ENCODES

Micro-aspiration of oropharyngeal secretions during sleep is the dominant route by which organisms reach the alveoli, even in healthy people. Impaired clearance or large inoculum tips to pneumonia.

BOARD TRAP

Micro-aspiration (silent, common) differs from gross aspiration pneumonia — the latter favors dependent segments and anaerobes/abscess.

3. Inhalation route

WHAT YOU SEE

'INHALE' on castle wall

WHAT IT ENCODES

Some pathogens reach the lung by direct inhalation of aerosols — classically atypicals (Legionella from water, Mycoplasma, viral). Droplet size determines alveolar deposition.

BOARD TRAP

Inhalation favors atypicals/Legionella (water exposure); micro-aspiration favors typical pneumococcus — exposure history guides empiric choice.

4. Mucosal barrier breach

WHAT YOU SEE

'MUCOSAL' on castle wall

WHAT IT ENCODES

The mucociliary escalator and mucosal immunity form the first defense. Viral infection, smoking, and COPD damage this barrier, predisposing to bacterial superinfection (e.g., post-influenza S. aureus/pneumococcus).

BOARD TRAP

Post-influenza pneumonia is classically S. aureus (including MRSA) and S. pneumoniae — a high-yield superinfection pairing.

5. Cytokine storm (IL-6, TNF, GM-CSF)

WHAT YOU SEE

Tornado of 'cytokine smoke', IL-6/TNF/GM-CSF vials

WHAT IT ENCODES

Pathogen recognition triggers IL-6, TNF, and GM-CSF release, driving fever, neutrophil recruitment, and the acute-phase response. CRP/procalcitonin rise. Excess cytokines cause systemic sepsis features.

BOARD TRAP

Procalcitonin elevation favors bacterial over viral pneumonia and can guide antibiotic stewardship, but should not solely dictate starting/stopping in severe illness.

6. Phase 1 — Edema (congestion)

WHAT YOU SEE

Room 1: yellow fluid, day 1

WHAT IT ENCODES

The first ~24h shows congestion: protein-rich edema fluid and bacteria fill alveoli with few neutrophils. The lobe is heavy and hyperemic. This is the earliest histologic phase of lobar pneumonia.

BOARD TRAP

Edema/congestion is bacteria-rich but neutrophil-poor — the inverse of red hepatization which is RBC/neutrophil-rich.

7. Phase 2 — Red hepatization

WHAT YOU SEE

Room 2: red 'LIVER' chamber

WHAT IT ENCODES

Days 2–3: alveoli pack with red cells, fibrin, and neutrophils, giving a firm liver-like consistency ('red hepatization'). Extravasated RBCs give the red color.

BOARD TRAP

Red hepatization = RBCs + fibrin + neutrophils; gray hepatization = RBCs disintegrated, fibrinopurulent. The hallmark is the loss of red color over time.

8. Phase 3 — Gray hepatization

WHAT YOU SEE

Room 3: gray 'CONTAINED' chamber

WHAT IT ENCODES

Days 4–6: red cells lyse and degrade, leaving a fibrinopurulent gray exudate; the lobe stays firm but turns gray. Macrophages begin to appear. Infection is being contained.

BOARD TRAP

Gray hepatization is dry/gray because RBCs have broken down — not because of new bacterial growth. Color change is a timing clue, not severity.

9. Phase 4 — Resolution

WHAT YOU SEE

Room 4: 'RESOLVED' open door, blue mop

WHAT IT ENCODES

Macrophages clear debris and the enzymatic digestion of exudate restores normal aerated architecture, typically over ~1 week onward. Lobar pneumonia characteristically resolves without scarring if uncomplicated.

BOARD TRAP

Lobar pneumonia resolves with restored architecture; persistent consolidation or scarring should prompt search for obstruction, abscess, or organizing pneumonia.

10. Lobar pattern (CXR)

WHAT YOU SEE

CXR gallery: 'LOBAR' film

WHAT IT ENCODES

Lobar consolidation — a dense opacity confined to one lobe with air bronchograms — is the classic pattern of *S. pneumoniae* and *Klebsiella*. Reflects alveolar filling.

BOARD TRAP

Lobar consolidation with a 'bulging fissure' classically suggests *Klebsiella*; pneumococcus is the commonest lobar cause overall.

11. Bronchopneumonia pattern

WHAT YOU SEE

CXR: 'BRONCHO' patchy opacities

WHAT IT ENCODES

Bronchopneumonia shows patchy, scattered, often bilateral opacities centered on airways — typical of *S. aureus*, Gram-negatives, and many nosocomial pathogens.

BOARD TRAP

Patchy multifocal pattern (bronchopneumonia) vs single dense lobe (lobar) helps narrow likely organisms — Staph/Gram-negatives vs pneumococcus.

12. Interstitial / atypical pattern

WHAT YOU SEE

CXR: 'ALVEOLAR' diffuse ground-glass, PJP inset

WHAT IT ENCODES

Diffuse ground-glass/interstitial infiltrates suggest atypical or viral pneumonia and *Pneumocystis jirovecii* (PJP) in immunocompromised hosts. CXR may look worse than the relatively mild exam in atypicals.

BOARD TRAP

Diffuse ground-glass in an HIV/immunosuppressed patient with hypoxia out of proportion = PJP — confirm with induced sputum/BAL and treat with TMP-SMX +/- steroids.

13. VAP / ventilator-associated

WHAT YOU SEE

CXR: 'VAP' tube, receding infiltrate

WHAT IT ENCODES

Ventilator-associated pneumonia (>48h after intubation) involves MDR organisms (*Pseudomonas*, MRSA, *Acinetobacter*). New/progressive infiltrate plus fever, leukocytosis, and purulent secretions define it.

BOARD TRAP

VAP requires broad empiric anti-pseudomonal/anti-MRSA cover; treating it like CAP is a major error. New infiltrate + clinical criteria, not infiltrate alone.

14. R→L shunt / hypoxemia

WHAT YOU SEE

Blue figure 'R→L' draining from castle

WHAT IT ENCODES

Consolidated alveoli are perfused but not ventilated, creating an intrapulmonary right-to-left shunt and hypoxemia that responds poorly to supplemental oxygen. V/Q mismatch worsens with filling.

BOARD TRAP

True shunt physiology (consolidation) corrects poorly with 100% O₂, unlike simple V/Q mismatch — a key gas-exchange distinction.

15. Trigger / inoculum

WHAT YOU SEE

Cannon 'TRIGGER' firing

WHAT IT ENCODES

Disease results when inoculum size and virulence overwhelm host defenses. Large aspirated bacterial load or impaired immunity (age, COPD, immunosuppression) converts colonization into invasive pneumonia.

BOARD TRAP

Host risk factors (age, COPD, immunosuppression, prior viral infection) determine progression more than mere exposure — risk-stratify before assuming severity.